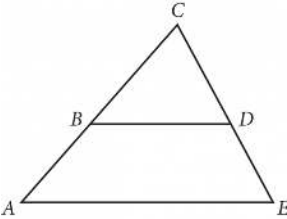


Question ID: 6dd463ca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure above, segments AE and BD are parallel. If angle BDC measures 58° and angle ACE measures 62° , what is the measure of angle CAE ?

Answer

- A. 58°
- B. 60°
- C. 62°
- D. 120°

Correct Answer: B

Rationale

Choice B is correct. It's given that angle ACE measures 62° . Since segments AE and BD are parallel, angles BDC and CEA are congruent. Therefore, angle CEA measures 58° . The sum of the measures of angles ACE, CEA, and CAE is 180° since the sum of the interior angles of triangle ACE is equal to 180° . Let the measure of angle CAE be x° . Therefore, $62 + 58 + x = 180$, which simplifies to $x = 60$. Thus, the measure of angle CAE is 60° .

Choice A is incorrect. This is the measure of angle AEC, not that of angle CAE. Choice C is incorrect. This is the measure of angle ACE, not that of CAE. Choice D is incorrect. This is the sum of the measures of angles ACE and CEA.